

Sagar Gupta

sg36.sagargupta26@gmail.com | +91-7872827780 | linkedin.com/in/sagar-gupta-02091783

Professional Summary

Backend Engineer with 6+ years of experience designing and building secure, scalable, high-availability backend systems using Java, Spring Boot, REST APIs, Microservices, and AWS. Experienced in authentication platforms, cloud migration, CI/CD modernization, and system modernization within regulated banking environments.

Technical Skills

Languages: Java 8, Java 17, JS(basic), Python(basic)

Frameworks and Technologies: Spring, Spring Boot, REST APIs, React

Architecture: Microservices, Cloud Native Architecture, Secure Authentication Systems, High Availability Systems

Databases: SQL (MySQL)

Cloud & DevOps: AWS, CI/CD, Git, GitLab CI, Jenkins, Docker(basic)

Engineering Practices: System Modernization, Cloud Migration, Mutation Testing, Technical Debt Reduction

Professional Experience

Barclays

Backend Engineer

Pune

Jul 2019 – Present

- Delivered app-based 2FA for Barclays Online Banking, replacing legacy PINsentry hardware. Improved authentication speed and user experience via push-based real-time approvals, delivering £6M savings over 5 years.
- Led CI/CD modernization by migrating build and deployment pipelines from Jenkins to GitLab CI, introducing automated quality, security, and compliance gates that improved release stability and developer productivity.
- Led modernization of legacy Java and Spring Boot services, improving application performance by 50% (3s-1.5s) while maintaining backward compatibility and supporting scalable microservices adoption.
- Implemented mutation testing using PIT (Java) and Stryker (React), strengthening test effectiveness and improving defect detection across backend and frontend applications.
- Created a separate journey to support accessibility and vulnerability requirements, enabling customers to receive bank statements in Braille, large print, and audio formats.
- Implemented a “Go Paperless” interstitial page prompting customers to switch from paper statements to digital delivery, reducing paper costs by £200K/year.
- Led cloud migration of authentication microservices to AWS, implementing cloud-native stateless services that improved scalability, high availability, and deployment reliability.
- Enabled reuse of an existing React application by exposing stable, backward-compatible backend APIs and integrating them into a mobile WebView flow.

TCG Digital

Mobile Developer Intern

Kolkata

Mar 2018 – May 2018

- Developed a mobile application using React Native to visualize import-export data across major Indian ports.

Academic & Personal Projects

- **New Product Sales Forecasting (Masters Project):** Implemented data pipelines and analytical models in Python to forecast product sales by combining historical data with sentiment analysis.
- **URL Shortener with Analytics:** Built a URL shortener with analytics using Spring Boot, Redis, and MySQL. Designed caching and lookup strategies achieving sub-50ms redirect latency and deployed on AWS EC2 with CloudWatch monitoring.

Education

Indian Institute of Technology, Kharagpur

Dual Degree (B.Tech + M.Tech), Industrial Engineering

CGPA: 7.39/10

2014 – 2019